

Article

Physiologically Explainable Ensemble Framework for Stress Classification via Respiratory Signals

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Abstract

This study proposes a physiologically interpretable framework for stress state classification using respiratory signals. The framework aims to assess whether integrating physiologically meaningful features with an interpretable model can enhance both the accuracy and interpretability of stress state classification. First, a 16-parameter feature set was constructed by extracting rhythm, depth, and nonlinear characteristics of respiratory signals. Subsequently, feature correlations and group differences across stress states were analyzed via heatmaps, multivariate analysis of variance (MANOVA), and box plots. A stacking ensemble model was then designed for three-state classification (normal/stress/meditation). Finally, Shapley additive explanations (SHAP) values were used to quantify feature contributions to classification outcomes. The leave-one-subject-out (LOSO) cross-validation results show that on the wearable stress and affect detection (WESAD) dataset, the model achieves an accuracy of 92.33% and a precision of 93.54%. Furthermore, initial validation shows key respiratory features like breath rate, inspiration time ratio, and expiratory variability coefficient align with autonomic regulation. Key respiratory metrics in other areas like rapid shallow breathing index also play an important role in the stress classification. Notably, increased respiratory depth under a stress state needs further study to clarify its physiological reasons. Overall, this framework enhances physiological interpretability while maintaining competitive performance, offering a promising approach for future applications in multimodal stress monitoring and clinical assessment.



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Keywords: respiratory signal; stacking; model interpretability; SHAP value; stress classification

1. Introduction

In the context of rapid modern life and escalating societal competition, there is a growing number of individuals experiencing psychological stress. The World Health Organization (WHO) reports 280 million global cases of depression, alongside rising anxiety disorder rates [1]. Accurate stress state identification is crucial for mental health evaluation and personalized intervention strategies. However, conventional methods of stress identification exhibit significant limitations. For instance, subjective scales like the perceived stress scale (PSS) stress perception scale can be affected by cognitive bias [2]. Behavioral observations, such as facial expression analysis, may lead to emotional concealment due to social expectations [3]. These limitations may constrain these approaches from offering comprehensive and objective assessments of underlying psychological states.

Recent advancements in physiological signal-based stress recognition address limitations of conventional methods. Stress responses correlate with sympathetic nervous

system (SNS) activation [4], measurable through heart rate [5], respiration [6], and skin conductance [6]. Notably, the respiration signal is distinguished by its dual characteristics. Firstly, the respiratory rhythm is bidirectionally modulated by the autonomic nervous system (ANS). Sympathetic dominance induces rapid shallow breathing linked to anxiety, while parasympathetic activity promotes slow breaths associated with relaxation [7]. Secondly, respiratory patterns are intrinsically coupled with limbic system activity [8], making them resistant to voluntary manipulation. Third, respiratory signals could be continuously recorded via non-invasive methods without electrode adhesion [9], offering higher feasibility compared to electroencephalogram (EEG) and electrocardiogram (ECG) for stress recognition [10]. These attributes position respiratory signals as promising biomarkers for objective stress assessment.

However, current exploration of respiratory signals for stress classification remains limited, with most studies addressing them only as part of multimodal signal analyses. For instance, S. Ghosh et al. achieved 94.77% accuracy in classifying four stress states by fusing six physiological signals (ECG, electromyography (EMG), respiratory (RESP), etc.) with deep learning [11]. Some studies even reached 99.7% accuracy in stress binary classification using multimodal signals, including respiratory ones [12]. However, these methods prioritize multimodal fusion over respiratory feature exploration and neglect evaluating the classification contribution of respiratory signals. Furthermore, some studies limit respiratory analysis to basic metrics like respiratory rate. For example, one approach involves predicting respiratory rate from multimodal signals and then using this derived parameter for stress prediction [13]. These results underscore the effectiveness of multimodal frameworks but reveal insufficient research on physiologically inspired respiratory features.

Current respiratory signal research mostly examines time domain and frequency domain statistical features [14–18]. Commonly employed features include respiratory rate, mean, variance, standard deviation, kurtosis, skewness, range (max-min), mean derivative, and spectral energy distribution across different frequency bands (e.g., 0.25–2.75 Hz) [6]. While these features capture signal variability, their highly mathematical abstraction prevents non-experts' intuitive understanding of personal breathing patterns. Critically, stress affects respiratory parameters in subtle ways, like altering inhalation/exhalation durations without changing respiratory rate [19,20]. In comparison, physiologically meaningful parameters, such as inhalation time (IT), exhalation time (ET), and the inhalation to exhalation ratio (I/E ratio), can capture stress-related information more precisely [21]. Physiologically inspired feature application in similar topics that involve respiratory analysis. For example, a higher I/E ratio links to relaxation and positive affect in emotion recognition. Moreover, key respiratory metrics like the rapid shallow breathing index (RSBI) are clinically relevant [22], especially in managing chronic obstructive pulmonary disease (COPD) and assessing acute heart failure (AHF). Previous studies also connect RSBI to anxiety, pain, and emotional states [22], highlighting its usefulness in stress characterization. Therefore, incorporating physiologically meaningful respiratory features may facilitate more intuitive and clinically actionable evaluations of stress states, particularly for use by healthcare professionals and end-users alike.

Moreover, model interpretability is essential for stress monitoring and management. Shapley additive explanations (SHAP) quantitatively identify the key features driving stress recognition, thereby informing targeted interventions grounded in these physiological determinants [23,24]. Although prior work has employed SHAP to elucidate decision processes in stress-identification models, the interpretability of respiratory features remains underexplored.

Therefore, we hypothesize that using physiologically meaningful respiratory features can enable effective and interpretable classification of stress states. To validate this, we propose an interpretable stress state recognition framework, validated using the wear-

able stress and affect detection (WESAD) dataset [25]. First, we evaluated the statistical properties of various physiologically meaningful features, including their correlations and intergroup differences across distinct stress states. Subsequently, a stacking ensemble model for three-state classification was developed to assess feature discriminative capacity. Finally, feature contributions were quantified using SHAP, establishing an interpretable decision pathway from physiological features to classification outcomes. This study preliminarily validates the potential of physiologically meaningful respiratory features in stress classification. Through an interpretable framework, we further analyze how these respiratory indicators influence the discrimination of distinct stress states.

The layout of this paper is as follows. Section 2 presents the dataset used in this study. Section 3 delineates the research methodology and technical route. Results are reported in Section 4, followed by discussions of the results in Section 5. Finally, we conclude the paper in Section 6.

2. Dataset

This study uses the WESAD dataset, a benchmark for emotion and stress research, published by a German team in 2018. It includes multimodal physiological signals from 15 subjects (12 males, 3 females, aged 24–35 years) recorded with a chest band (RespiBAN, Plux Wireless Biosignals S. A., Lisboa, Portugal) and wrist-worn (Empatica E4, Empatica, Boston, MA, USA) device. The subjects did not include pregnant individuals, heavy smokers, or those with psychiatric disorders, chronic illnesses, or cardiovascular diseases. All participants were instructed to avoid caffeine and tobacco for at least one hour prior to the experiment and to refrain from vigorous physical activity on the day of testing. The RespiBAN recorded ACC, ECG, EDA, EMG, RESP, and TEMP. The Empatica E4 captured ACC, BVP, EDA, and TEMP. Detailed descriptions of the experimental conditions, controlled variables, and study protocol can be found in the original dataset publication [25].

The dataset included four states: baseline (resting reading), stress (via the Trier Social Stress Test [26]), amusement (self-selected activities), and meditation (mindful rest). We focused on respiratory signals from the chest-band sensor, sampled at 700 Hz. Participants underwent the four scenarios sequentially with continuous respiratory data collection.

Current WESAD-based studies mostly use multimodal signals for binary or ternary stress classification [27], as WESAD is a widely recognized dataset for evaluating stress detection performance. In contrast, our study focuses on exploring the potential of physiological interpretability in stress classification using respiratory signals alone. For real-world applications like breathing-guided interventions, explicit meditation state classification is critical, as it shows stress relief outcomes. Baseline and amusement states were merged into a unified normal state. This 3-class framework (stress/normal/meditation) meets practical stress management needs and supports intervention decisions. To match the research objectives, the original WESAD dataset labels were redefined, merging baseline and amusement states into one normal state. The specific allocation per subject was 24 normal state samples, 10 stress state samples, and 12 meditation state samples, as detailed in Section 3.

3. Materials and Methods

Signal preprocessing is initially accomplished through filtering and periodic segmentation. This is followed by the extraction and statistical analysis of respiratory rhythm, depth, and nonlinear dynamical features. Subsequently, a stacking ensemble learning model is constructed to ascertain its efficacy in classifying stress states. Finally, feature contributions were quantified using SHAP. The overall framework is shown in Figure 1.

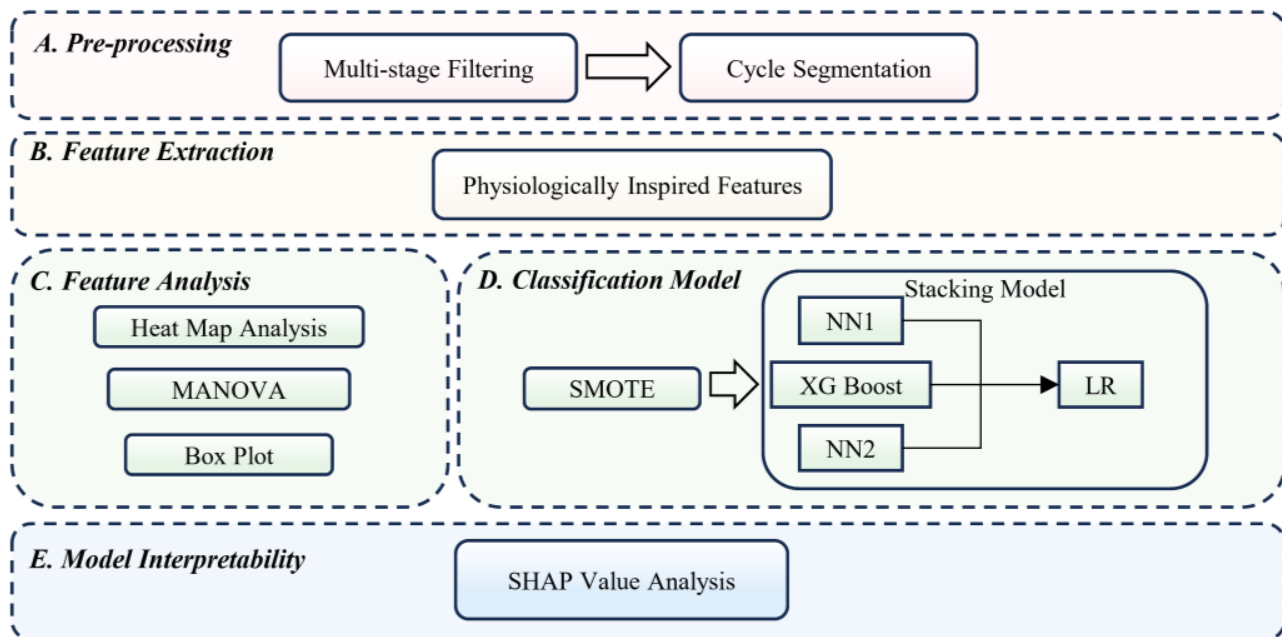


Figure 1. Overall framework of the research method.

3.1. Respiratory Signal Preprocessing

The preprocessing of respiratory signals encompasses two stages: multi-level noise suppression and respiratory cycle segmentation. This process is illustrated in Figure 2. Initially, apply an 8th-order Butterworth low-pass filter (80 Hz cutoff) to the original signal to remove electromyography artifacts and high-frequency noise. Subsequently, use a 700-sample point sliding window moving average to reduce short-term fluctuations and motion artifacts.

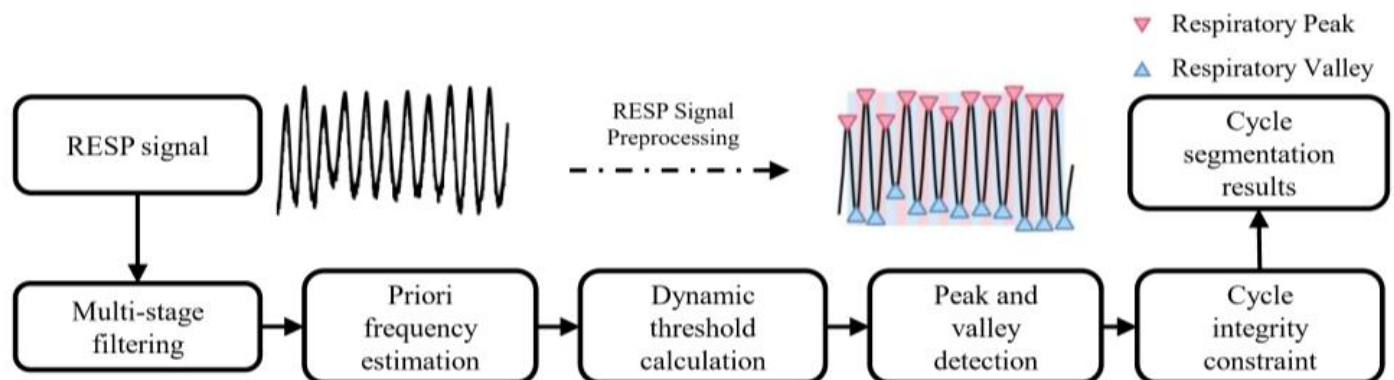


Figure 2. Flow chart for preprocessing.

To delineate the respiratory cycle, identifying local maxima (indicating inspiration termination) for peak point extraction is essential. However, conventional methods may inaccurately detect transient pseudo-peaks caused by respiratory dynamic disruptions. These pseudo-peaks, appearing as isolated waveforms in the respiratory rhythm, differ from true expiratory termination morphology. Thus, a dynamic threshold adjustment strategy is adopted to avoid spurious peak detection and prevent erroneous segmentation. The specific procedure is as follows:

Prior frequency estimation: Use the fast Fourier transform (FFT) to identify the fundamental frequency (Fres) corresponding to the peak energy in the 0.1–0.5 Hz band.

Dynamic threshold calculation: Compute the time-varying threshold by combining global statistical attributes (mean μ and standard deviation σ) with local sliding windows. The window length is $W = 2 F_s / F_{res}$, with F_s at 700 Hz.

Peak and valley detection: Apply a dynamic threshold peak detection algorithm to locate and record the exact positions of peaks and valleys in the respiratory signal.

Cycle integrity constraint: Chronologically organizes the detected peaks and troughs. Remove consecutive identical peak or trough types to ensure alternating peaks and troughs, maintaining the integrity of the reconstructed sequences.

This processing flow effectively resolves mis-segmentation issues in traditional fixed threshold methods during dynamic breathing, while preserving breathing waveform morphology.

3.2. Respiratory Signal Feature Extraction

This study uses non-overlapping one-minute windows on preprocessed respiratory signals. Sixteen features from three categories are extracted based on respiratory cycle segmentation. Feature extraction follows the respiratory pattern quantification analysis framework [28,29], covering rhythm, depth, and nonlinear dynamical parameters. A comprehensive summary of the features is provided in Table 1. The definitions and calculation methods of each feature are as follows.

Table 1. Classification and definition of respiratory signal features used in this study.

Feature Category	Parameter Name	Description
Respiratory Rhythm Parameters	BR	Breathing rate mean (bpm)
	BR_cv	Respiratory rate variation coefficient, refer to Formula (1)
	IT	Inspiration time mean, mean time interval from trough to next peak (s)
	IT_cv	Inspiratory time variation coefficient, refer to Formula (1)
	ET	Expiratory time mean, mean time interval from peak to next trough (s)
	ET_cv	Expiratory time variation coefficient, refer to Formula (1)
	IT_ratio	Inspiratory time ratio, refer to Formula (2)
	IT_ratio_cv	Inspiratory time ratio variation coefficient, refer to Formula (1)
Respiratory Depth Parameters	D	Breathing depth mean, the mean difference between the amplitude of a wave crest and an adjacent trough
	D_cv	Breathing depth ratio variation coefficient, refer to Formula (1)
	RSBI	Rapid shallow breathing index, refer to Formula (3)
	RSBI_cv	Rapid shallow breathing index variation coefficient, refer to Formula (1)
Respiratory Nonlinear Dynamics Parameters	SD1	Short-Term Variability of Poincaré chart, Formula (4)
	SD2	Long-Term Variability of Poincaré chart, Formula (5)
	Appor	Approximate Entropy, calculation method refer to [30]
	Sample	Sample Entropy, calculation method refer to [30]

3.2.1. Respiratory Rhythm Feature

The measurement of a single breathing cycle, defined as the time interval between adjacent waves (denoted as BB), enables the calculation of the breathing rate. This rate,

when averaged over a one-minute period, yields the averaged breathing rate (BR, unit: times/min). The calculation of the breathing rate coefficient of variation (BR_cv) is (1).

$$BR_cv = \frac{\sigma_{BRi}}{\mu_{BRi}} \quad (1)$$

where μ is the mean, and σ is the standard deviation. In a similar vein, both the mean values and coefficients of variation are calculated for respiratory rhythm characteristics and respiratory depth parameters.

The inspiration time (IT) is delineated as the temporal interval spanning from a trough to its subsequent peak, while the expiration time (ET) is characterized by the interval between a peak and the ensuing trough. The calculation of the inspiration time ratio (IT_ratio) is delineated in (2) as the ratio of the inspiration time to the duration of the entire respiratory cycle. This ratio provides insights into the asymmetry of the various respiratory phases.

$$IT_ratio_i = \frac{IT_i}{IT_i + ET_i} \quad (2)$$

3.2.2. Respiratory Depth Feature

The respiratory depth (D) is defined as the differential in amplitude between the peaks and troughs of the respiratory waveform. This measurement does not have a specific unit. The method for calculating the coefficient of variation of respiratory amplitude is the same as previously described. Equation (3) is used to compute the rapid shallow breathing index (RSBI), which is the ratio of respiration rate to respiratory amplitude in a single cycle and reflects the respiratory compensation mode.

$$RSBI_i = \frac{BR_i}{D_i} \quad (3)$$

3.2.3. Nonlinear Dynamic Feature

The poincaré plot (PP) is a frequently employed nonlinear indicator [31]. As defined by the methodology in [28], the short axis (SD1) shows short-term complexity via immediate respiratory interval variability fluctuations. The long axis (SD2) indicates overall stability by representing the enduring trend of respiratory interval variability.

$$SD1 = \frac{\sqrt{2}}{2} \sigma(\Delta BB_i - \Delta BB_{i+1}) \quad (4)$$

$$SD2 = \sqrt{2\sigma^2(BB_i) - SD1^2} \quad (5)$$

Approximate entropy (Appro) and sample entropy (Sample) are used to quantify respiratory rhythm irregularity because they effectively measure the complexity and unpredictability of time series data [30]. Approximate entropy reflects a sequence's self-similarity and is suitable for non-stationary and nonlinear data. Sample entropy, an improved version, measures the probability of new patterns emerging in a time series, making it more reliable for assessing physiological signals like respiratory rhythms. Both methods are widely used in physiological time series analysis and can reveal respiratory pattern alterations during stress. When calculating the entropy, the parameters are set as follows: the embedding dimension m is set to 2, and the similarity tolerance r is set to 2 times the signal standard deviation σ .

To validate the features we used for stress classification, we built a comparative framework using general respiratory signal features. General stress classification methods using respiration typically involve time and frequency domain analysis, extracting statistical

features from both time and frequency domain analyses. In this study, we selected standard time and frequency respiratory features from the raw signal and added wavelet domain statistical features to form the baseline feature set. This allows a quantitative performance comparison, confirming the superior discriminatory power of our new feature sets for stress classification. Details of general respiratory features are in Table 2.

Table 2. Classification and definition of general respiratory signal features.

Feature Category	Parameter Name	Description
Time Domain Feature Parameters	Avg, Std, Rat, Skw, Kur	Mean, standard deviation, ratio of maximum value to mean, skewness, and kurtosis.
Frequency Domain Feature Parameters	Fre, Fre1, Fre2, Fre3, Fre4, Fre5, Fre6	Main frequency, power sum in 0–0.1 Hz, 0.1–0.2 Hz, 0.2–0.3 Hz, 0.3–0.4 Hz, 0.4–0.7 Hz, and 0.7–1 Hz bands.
Wavelet Domain Feature Parameters	Wer, Wee, We, Wse	Wavelet energy ratio of the first subband, wavelet energy entropy, wavelet entropy of the first subband, and wavelet singular entropy.

3.3. Respiratory Signal Feature Analysis

To explore correlations among respiratory signal feature categories and their ability to characterize respiratory information, this study used heatmap analysis to quantify the Pearson correlation coefficient matrix [32]. This revealed correlation patterns among respiratory features across emotional states. Coefficients represent pairwise feature correlations, visualized via a color gradient where darker colors indicate stronger correlations. This method aids in understanding relationships across respiratory feature categories, supports feature selection, and enhances model interpretability.

Multivariate analysis of variance (MANOVA) is a statistical method for analyzing differences in multiple dependent variables among two or more groups [33]. It assesses whether groups differ significantly on a combination of dependent variables, accounting for their correlations. In this study, MANOVA was employed to assess the between-group differences and quantify the multivariate effect of stress state on respiratory features. The computed features served as dependent variables, with stress state (normal/stress/meditation) as the fixed factor. The Pillai's Trace statistic was used as the primary criterion for significance, with a threshold set at $\alpha = 0.05$. Upon the MANOVA revealed a significant main effect ($p < 0.05$), subsequent pairwise comparisons were conducted utilizing Tamhane's T2 test, which accounts for the heterogeneity of variance. This approach pinpointed specific respiratory parameters that exhibited significant differences between groups while maintaining control over the type I error rate.

To better visualize respiratory pattern differences across stress states, this study uses box plots to display features in three stress states. To unify dimensions and compare distribution differences of respiratory features, Z-score normalization is applied to all features. The box plots depict the medians, the 25th and 75th percentiles, extreme values corresponding to 1.5 times the interquartile range (IQR) indicated by whiskers, and outliers represented as discrete points for each feature.

3.4. Stress Classification Model

First, the data were baseline-removed to enhance generalization ability. Then, to address class imbalance in stress state classification, the synthetic minority over-sampling technique (SMOTE) was applied to oversample and balance the stress and meditation

classes [34], ensuring equal sample sizes and boosting model reliability. During baseline removal, subject-specific baselines were computed as the mean of each feature across states and subtracted from the raw data. To counter class imbalance, the stress and meditation state samples were oversampled via SMOTE until all three classes matched the sample size of the normal state. Our validation framework employs a stacking model. The stacking model combines predictions from multiple base models to improve generalization and reduce overfitting risk.

The stacking model uses three base learners:

- A Neural Network 1 with 64, 64 neurons in hidden layers, ReLU activation, Adam solver, regularization ($\alpha = 0.0001$), and a maximal number of iterations of 50.
- A Neural Network 2 with 32, 64, 128 neurons in hidden layers, ReLU activation, Adam solver, regularization ($\alpha = 0.03$), and a maximal number of iterations of 50.
- A Gradient Boosting with Extreme Gradient Boosting Random Forest method, 50 trees, learning rate 0.1, regularization lambda 0.001, limit depth of individual trees 10, and replicable training.

The meta-learner is a logistic regression model for final prediction integration, and the Logistic Regression with Ridge (L2) regularization, $C = 1$, and no balanced class distribution.

This study used data from 15 subjects. Data were balanced with SMOTE, giving 72 samples per subject (1080 total). LOSO cross-validation was applied. In each fold, the training set comprised class-balanced data from 14 subjects (1008 samples); the test set consisted of the remaining subjects' data, which retained its original distribution and underwent only baseline removal. After testing all 15 subjects, the average performance across these tests was taken as the final result. Evaluation metrics were calculated as follows [35]:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (6)$$

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (7)$$

where Precision = $\text{TP}/(\text{TP} + \text{FP})$, Recall = $\text{TP}/(\text{TP} + \text{FN})$. Accuracy measures correct predictions over all samples, showing overall classification accuracy. The F1 score, the harmonic mean of precision and recall, effectively evaluates models in class-imbalance cases. Higher values of these two metrics mean better model performance.

3.5. Classification Model Interpretability

This study clarifies the significance of individual characteristics in differentiating emotional states through a classification model. It uses SHAP values to evaluate the model's interpretability [36]. The SHAP value of the j -th feature in the i -th sample quantifies its contribution to the model's prediction for that sample, while also providing insights into feature importance and model behavior.

$$\phi_j^{(i)} = \sum_{S \subseteq \mathcal{P} \setminus \{j\}} \frac{|S|!(|\mathcal{P}| - |S| - 1)!}{|\mathcal{P}|!} [f(S \cup \{j\}) - f(S)] \quad (8)$$

where $\mathcal{P}$ is the complete set of features, and $f(S)$ represents the model prediction output using the feature subset S . The SHAP absolute mean $|\bar{\phi}_j|$ represents the global feature importance.

Signal preprocessing and feature extraction were performed using MATLAB 2019b. Origin 2025 was used for feature correlation analysis and heatmap generation. MANOVA statistical analysis and box plot creation were done using SPSS 27.0. All model building, training, and testing were conducted on the open-source Orange 3.8 platform, where SHAP values were calculated. Predictive SHAP value plots were created using Python 3.10.

4. Results

4.1. Respiratory Signal Feature Analysis Results

Figure 3 shows a correlation heatmap of respiratory parameters, revealing low correlations ($|r| < 0.45$) among respiratory rhythm, amplitude, and nonlinear dynamics. This indicates that the multi-dimensional feature space can effectively distinguish different physiological responses to stress by capturing various aspects of respiratory dynamics. The low correlation between features ensures each dimension contributes unique information, which is essential for building robust and interpretable models.

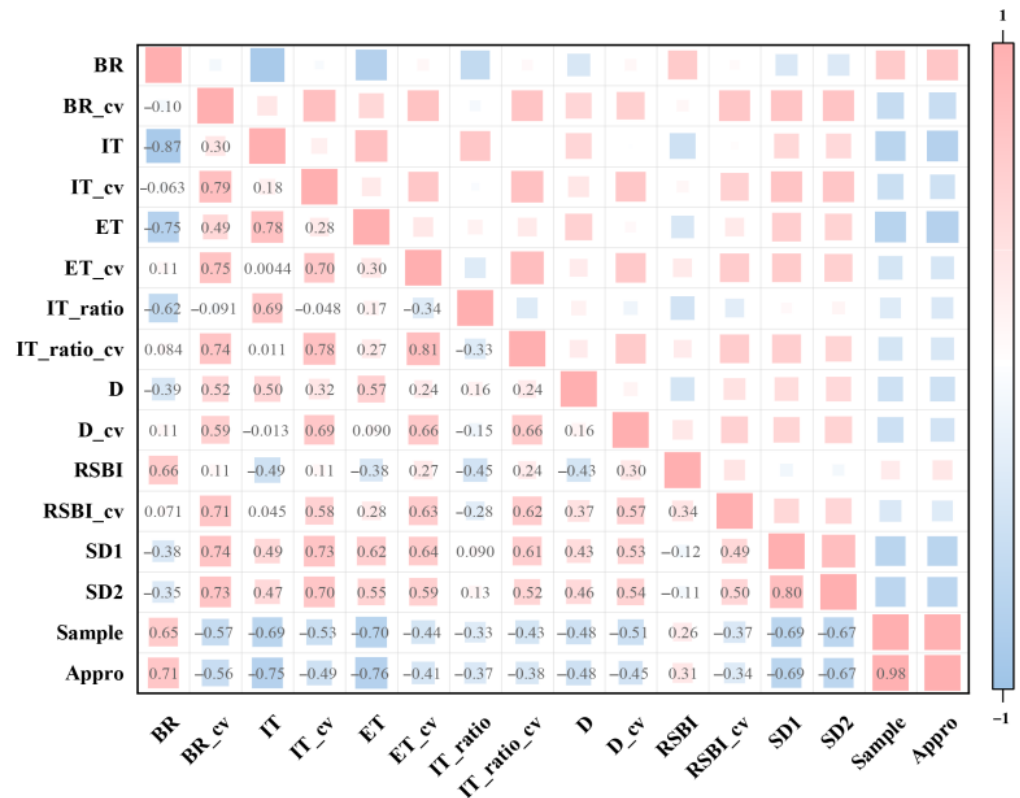


Figure 3. Correlations among different respiratory features.

Table 3 offers statistical comparisons of respiratory features across three stress states (N, S, M). Most parameters show significant differences ($p < 0.05$). Compared to the normal state, stress conditions induce distinct respiratory alterations: a substantial elevation in breathing rate (BR: $\Delta M = +1.6281$, $p < 0.001$) with increased variability (BR_cv: $\Delta M = +0.2088$, $p < 0.001$), concurrent with reduced inspiratory time (IT: $\Delta M = -0.1416$, $p = 0.002$) and heightened expiratory variability (ET_cv: $\Delta M = +0.2561$, $p < 0.001$). In contrast, meditation exhibits an opposing regulatory pattern, marked by significant BR suppression (BR: $\Delta M = -5.8530$, $p < 0.001$), prolonged inspiratory duration (IT: $\Delta M = +1.7060$, $p < 0.001$), and enhanced inspiratory dominance (IT_ratio: $\Delta M = +0.048$, $p < 0.001$).

Table 3. Post-hoc multiple comparison results between groups.

Feature	I	J	I – J	Error	Sig.	Feature	I	J	I – J	Error	Sig.
BR	N	S	−1.6281 *	0.2974	0.000	BR_cv	N	S	−0.2088 *	0.0095	0.000
	S	M	7.4811 *	0.3347	0.000		S	M	0.0675 *	0.0138	0.000
	M	N	−5.8530 *	0.2291	0.000		M	N	0.1413 *	0.0122	0.000
IT	N	S	0.1416 *	0.0412	0.002	IT_cv	N	S	−0.1865 *	0.0115	0.000
	S	M	−1.8480 *	0.0811	0.000		S	M	0.0920 *	0.0146	0.000
	M	N	1.7063 *	0.0749	0.000		M	N	0.0945 *	0.0123	0.000
ET	N	S	−0.2933 *	0.0324	0.000	ET_cv	N	S	−0.2561 *	0.0089	0.000
	S	M	−0.5970 *	0.0582	0.000		S	M	0.1647 *	0.0118	0.000
	M	N	0.8903 *	0.0506	0.000		M	N	0.0914 *	0.0104	0.000
IT_ratio	N	S	0.0506 *	0.0034	0.000	IT_ratio_cv	N	S	−0.1413 *	0.0064	0.000
	S	M	−0.0990 *	0.0039	0.000		S	M	0.1014 *	0.0079	0.000
	M	N	0.0484 *	0.0034	0.000		M	N	0.0399 *	0.0064	0.000
D	N	S	−2.5176 *	0.2909	0.000	D_cv	N	S	−0.2124 *	0.0157	0.000
	S	M	−0.4953	0.4778	0.658		S	M	0.1559 *	0.0198	0.000
	M	N	3.0129 *	0.4036	0.000		M	N	0.0565 *	0.0200	0.015
RSBI	N	S	−1.8441 *	0.4118	0.000	RSBI_cv	N	S	−0.7210 *	0.0452	0.000
	S	M	3.6690 *	0.4193	0.000		S	M	0.5091 *	0.0545	0.000
	M	N	−1.8249 *	0.2101	0.000		M	N	0.2119 *	0.0448	0.000
SD1	N	S	−0.7652 *	0.0458	0.000	SD2	N	S	−0.6454 *	0.0432	0.000
	S	M	−0.1383	0.0831	0.264		S	M	−0.1675	0.0712	0.057
	M	N	0.9035 *	0.0767	0.000		M	N	0.8129 *	0.0665	0.000
Sample	N	S	0.0009 *	0.0001	0.000	Appro	N	S	0.0008 *	0.0001	0.000
	S	M	0.0012 *	0.0001	0.000		S	M	0.0015 *	0.0001	0.000
	M	N	−0.0022 *	0.0001	0.000		M	N	−0.0023 *	0.0001	0.000

Note: I and J represent different stress state groups, N: normal, S: stress, M: meditation; Sig. stands for significance. *: indicates that the difference between the data is statistically significant.

Nonlinear dynamics analysis reveals stress-induced amplification of short-term respiratory rhythm variability through increased Poincaré plot parameter SD1 (SD1: $\Delta M = +0.7652$, $p < 0.001$). Conversely, meditation manifests respiratory regularization, evidenced by reduced signal complexity through decreased sample entropy (Sample: $\Delta M = -0.0022$, $p < 0.001$) and approximate entropy (Appro: $\Delta M = -0.0023$, $p < 0.001$). These findings systematically quantify emotion-specific respiratory modulation patterns across temporal, variability, and nonlinear domains.

Figure 4 presents boxplot visualizations comparing respiratory parameter distributions across three states after analyzing feature relationships and group-wise statistical differences. In the stress state, respiratory acceleration is evident, shown by a higher BR median and wider IQR, indicating increased variability. In contrast, meditation is marked by respiratory stabilization, with lower RSBI values and a narrower IQR, reflecting more regular breathing patterns. Normal state parameters exhibit narrow, symmetrical distributions, confirming baseline respiratory consistency. These positional and dispersion characteristics support the statistical tests, confirming stress-specific respiratory signatures across temporal and variability domains.

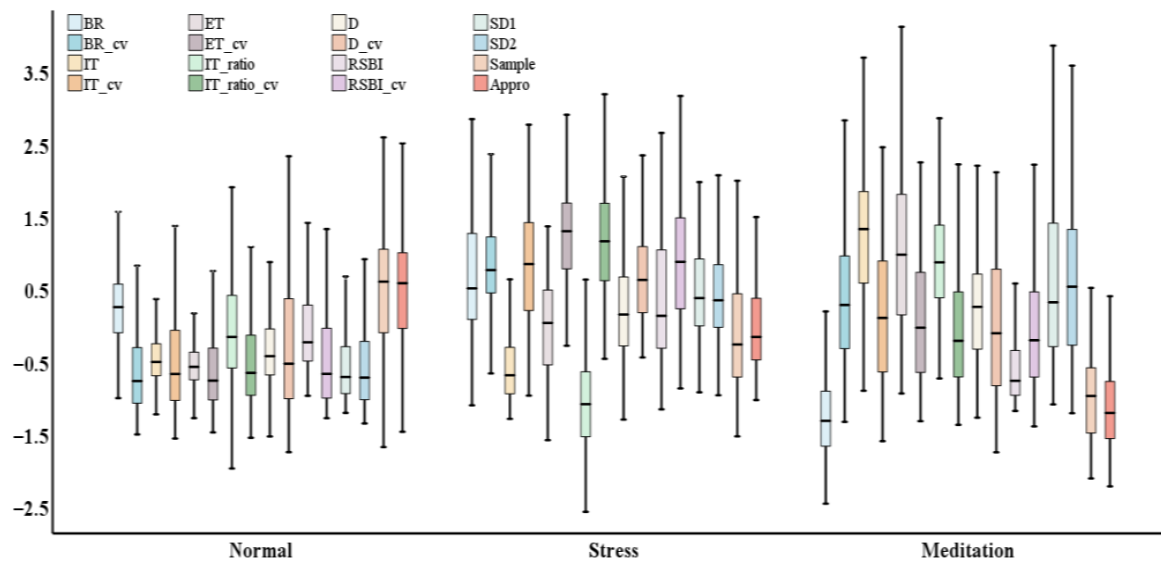


Figure 4. Feature boxplots of different stress states.

4.2. Stress Classification Performance

Table 4 summarizes the LOSO cross-validation results of stress classification tests and overall performance. The framework integrates newly extracted respiratory features with a stacking model, resulting in effective classification performance. Overall, it reaches 92.33% accuracy, 92.32% F1 score, 93.54% precision, and 92.33% recall. On an individual level, most subjects have over 90% accuracy and F1 scores, with significant consistency in F1 and accuracy across models. Though a few individuals have lower accuracy, subsequent in-depth analysis can explore the reasons. Generally, the experiment confirms the robust representational capability of respiratory features for stress state classification, achieving 92.33% accuracy.

Table 4. Experimental results of stress state classification.

Subject	Accuracy	F1	Precision	Recall
2	91.3%	91.0%	91.6%	91.3%
3	100.0%	100.0%	100.0%	100.0%
4	84.8%	85.3%	87.5%	84.8%
5	93.5%	93.3%	93.8%	93.5%
6	95.7%	95.6%	96.0%	95.7%
7	100.0%	100.0%	100.0%	100.0%
8	93.5%	93.6%	93.8%	93.5%
9	97.8%	97.8%	98.0%	97.8%
10	91.3%	91.3%	91.8%	91.3%
11	87.0%	85.7%	88.7%	87.0%
13	67.4%	68.7%	76.5%	67.4%
14	89.1%	89.2%	91.5%	89.1%
15	95.7%	95.5%	96.0%	95.7%
16	97.8%	97.8%	97.9%	97.8%
17	100.0%	100.0%	100.0%	100.0%
Average	92.33%	92.32%	93.54%	92.33%

The confusion matrix in Figure 5 reveals that the normal state is most often misclassified as the stress state. This is due to extreme normal state samples, such as those with high breathing rates and unstable patterns, being mistakenly identified as stress. However, there is little confusion between stress and meditation states. This indicates their distinct breathing patterns are not easily confused.

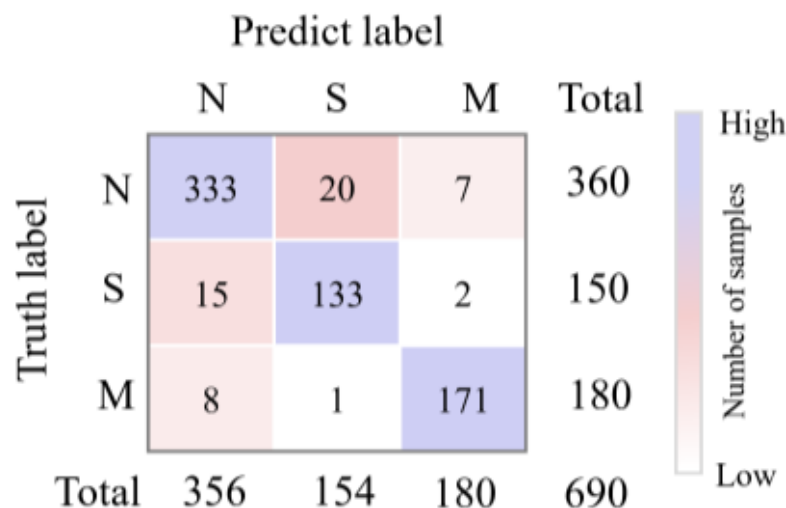


Figure 5. The confusion matrix generated when utilizing this feature in conjunction with the stacking model is presented herein.

Table 5 presents the stacking model's performance in three stress classification scenarios. The model best identifies the meditation state, with 97.39% accuracy and 95.00% F1 score. In the normal state, the model shows a balanced performance with an F1 score of 93.02%. However, the stress state classification accuracy was 94.49%, and the F1 score was 87.50%. According to the confusion matrix, the stress state was often confused with the normal state, resulting in lower precision and recall. The differences in accuracy across categories are probably due to the varying distinguishability levels in breathing patterns.

Table 5. Performance of the stacking model in three categories of stress classification.

State	Accuracy	F1	Precision	Recall
Normal	92.75%	93.02%	93.54%	92.50%
Stress	94.49%	87.50%	86.36%	88.67%
Meditation	97.39%	95.00%	95.00%	95.00%

Table 6 presents the results of different feature tests. The proposed feature set consists of 16 physiologically meaningful respiratory features, while the general respiratory features set also contains 16 commonly used respiratory features. Details of general respiratory features are in Table 2. A hybrid feature set is constructed by combining both, resulting in a total of 32 features. The proposed feature combination achieved the highest values across all four metrics, with a 10.58% improvement in accuracy and a 10.69% increase in F1 score over generic features. When hybrid features were used, the accuracy reached 90.73%, a significant boost from general features but still lower than the proposed ones. These enhancements in accuracy and F1 score indicate that our features capture more discriminative information, which is highly valuable for inter-individual stress classification.

Table 6. Results of the stress classification test based on different features.

Feature Set	No. of Features	Accuracy	F1	Precision	Recall
This Study	16	92.33%	92.32%	93.54%	92.33%
General	16	81.75%	81.63%	84.50%	81.75%
Hybrid	32	90.74%	90.79%	92.27%	90.74%

4.3. SHAP Value Analysis Result

In the normal state, the model is heavily influenced by BR, ET_cv, and IT_ratio. An increase in ET_cv negatively affects the model, while a higher IT_ratio positively affects normal state recognition (Figure 6). In the stress state, the coefficient of variation of related features has a bigger impact. The effects of IT_ratio and ET_cv reverse from the normal state. This aligns with prior analysis, indicating stress leads to reduced IT_ratio and increased breathing pattern variability. In the meditation state, BR's influence on the model is opposite to that in normal and stress states. Higher BR negatively impacts meditation state judgment. The analysis also shows that IT_ratio and IT significantly and positively affect meditation state judgment. This makes sense as meditation is linked to reduced BR and extended IT. Overall, the stress classification rules based on feature values align with the feature distribution rules in different stress states from our previous statistical analysis.

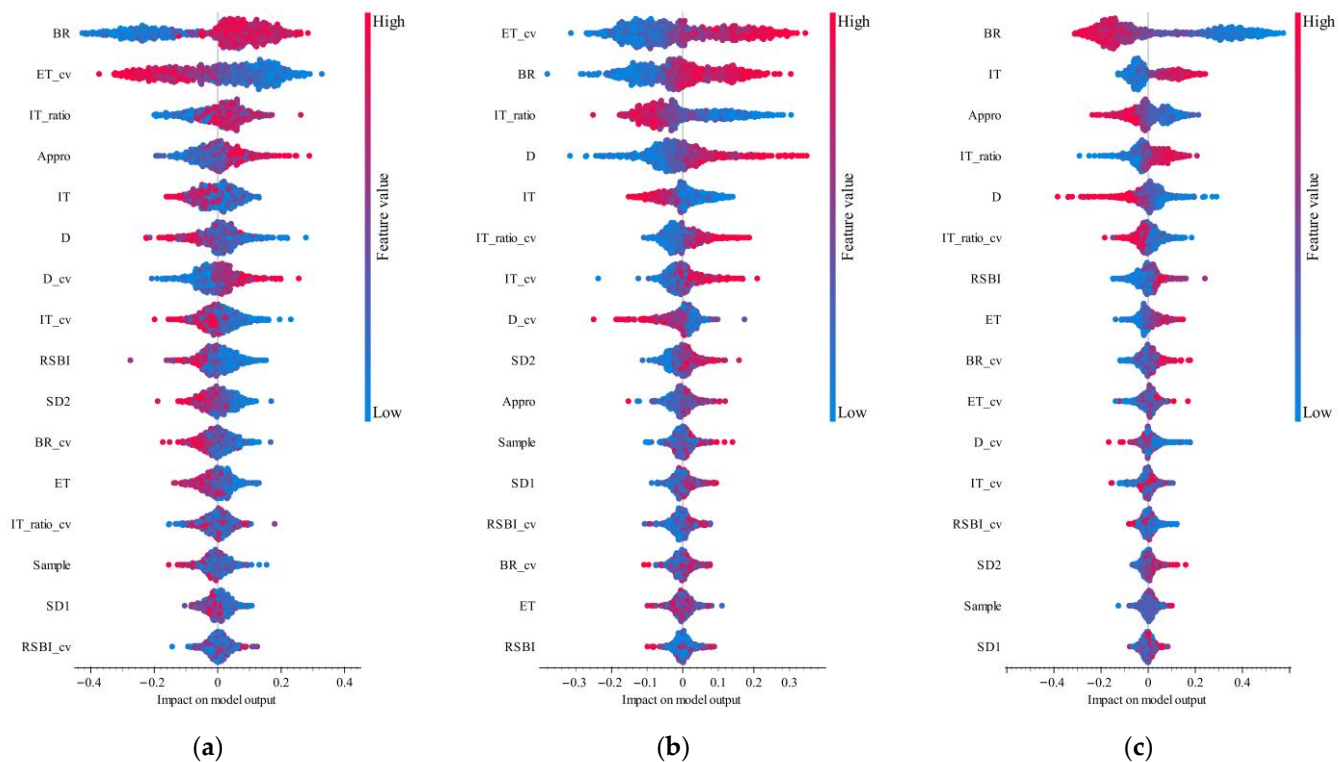
**Figure 6.** (a) SHAP values of features in the normal state, (b) SHAP values of features in the stress state, (c) SHAP values of features in the meditation state.

Figure 7 compares the actual and predicted states of Subject 16. The analysis identified only one inaccuracy: one instance where a stress state was misclassified as normal. Figure 8 uses SHAP values to quantify the contribution of features to the final model prediction. Our analysis identifies RSBI_cv and BR_cv as key determinants of model predictive performance. Their absolute values significantly affect output probabilities. While feature importance distributions vary across individual samples, those with the same predicted state share

many similarities. This indicates intra-state consistency in respiratory dynamics, with minimal feature value variability accompanying physiological homogeneity within each stress state.

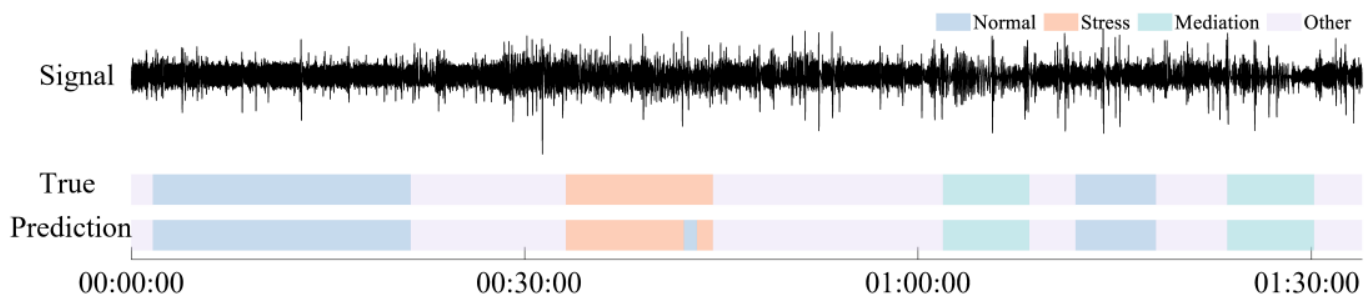


Figure 7. Comparison diagram of the actual state and predicted state for Subject 16.

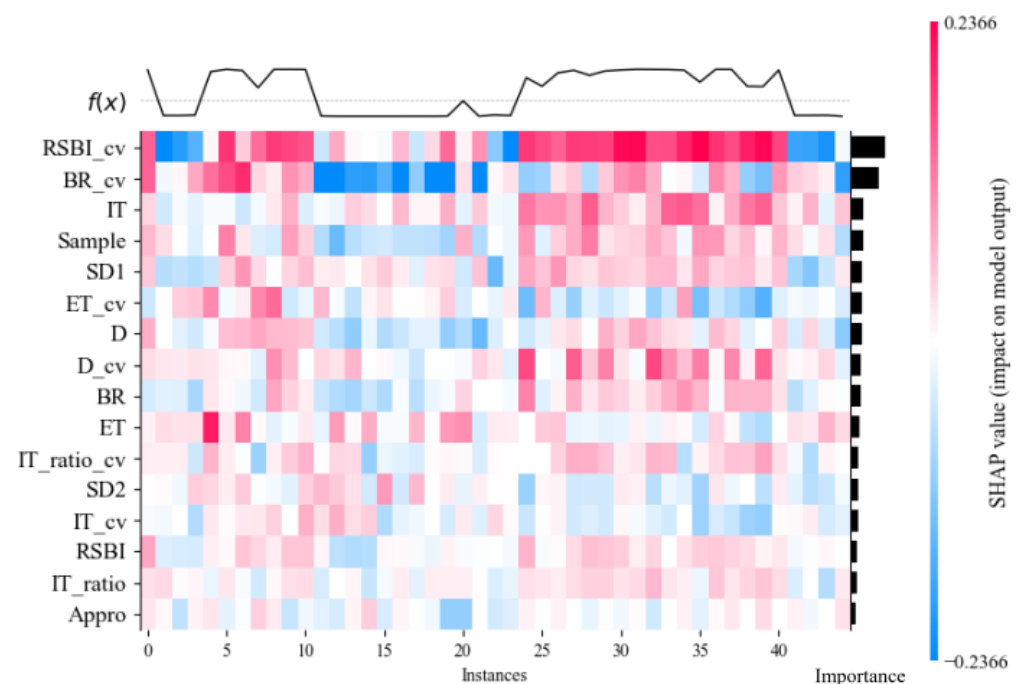


Figure 8. Feature SHAP values yielded at the time of model prediction.

5. Discussion

5.1. Respiratory Feature Analysis

Our research introduces a new respiratory pattern parameter quantification analysis technique, creating a multi-dimensional feature space covering respiratory rhythm, depth, and nonlinear dynamical characteristics. This is the first application of this quantification approach to stress state recognition. The classification results preliminarily confirm that physiologically inspired features have the potential to distinguish between different stress states.

MANOVA and boxplot analyses found significant differences in core parameters like BR and IT and their variability coefficients. Under stress, BR increases with higher variability, indicating rapid and unstable respiration. Meditation reduces BR and ET_cv, showing restored respiratory regularity and autonomic regulation. Additionally, meditation significantly increased IT_ratio, potentially linked to breath-holding states during inhalation in meditation, prolonging inspiration (Figure 9).

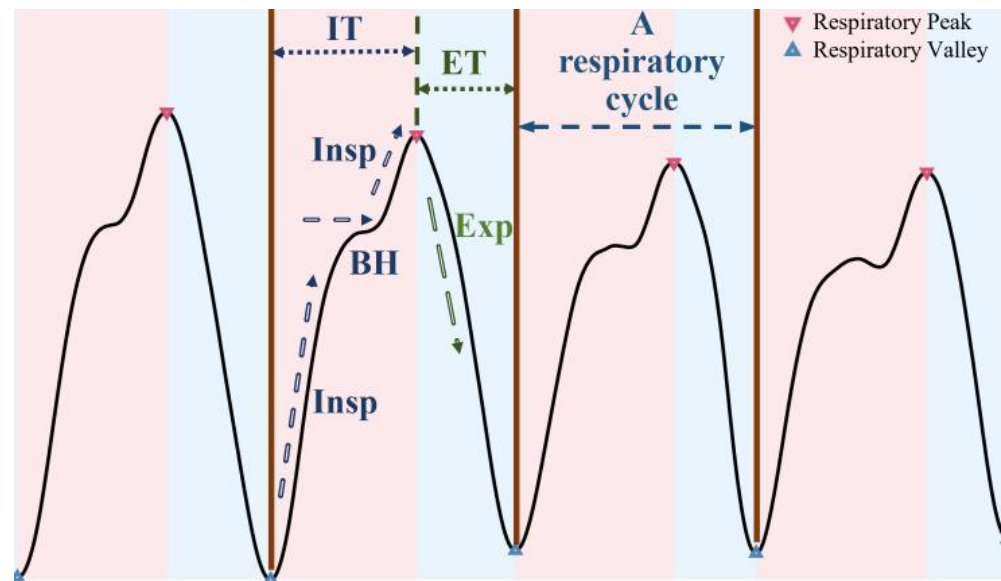


Figure 9. Respiratory waveform during meditation state.

Interestingly, D was found to be greater under stress than in the normal state, which might be due to stress-induced deep breathing to enhance oxygen intake and relieve tension. This could also reflect dataset bias, as the study group was predominantly male, who typically have deeper breathing patterns. Additionally, SD1 and SD2 features showed no significant differences between stress and meditation states, suggesting they have limited use in assessing short-term versus long-term respiratory stability.

5.2. Model Classification Performance

This study classified stress considering the full cycle of actual stress regulation, with dataset labels divided into normal, stress, and meditation states. This classification helps assess meditation intervention outcomes and develop personalized stress management strategies. To the best of our knowledge, this is the first study to integrate physiologically meaningful respiratory features with an interpretable ensemble model for stress classification. Our labeling method differs from traditional WESAD dataset methods, so we tested its efficacy using general features. The results showed our features outperformed general ones, improving accuracy by 10.58%, highlighting our method's effectiveness.

As Table 7 shows, most WESAD-based studies utilize multimodal data to improve accuracy, but such approaches may increase computational complexity and limit deployment in embedded systems. In contrast, our study focuses on respiratory signals and adopts a three-class classification framework (normal/stress/meditation) by redefining the original labels. Since several WESAD-based studies omit the meditation state, comparisons mainly focus on stress and related state accuracy. For instance, study [11] reported a meditation accuracy of 94.55%, while our study achieved a higher accuracy of 97.39%. Our stress recognition accuracy reached 94.49%, slightly below the 95.21% reported in [27] for binary stress classification. Notably, our accuracy exceeds the 84.32% reported in [27] for three-class classification. Though slightly lower than some multimodal or ECG-based methods [13,37], our approach achieves comparable results using only respiratory signals. It is important to note, however, that differences in validation protocols may significantly influence the performance outcomes.

Table 7. Summary and comparison of related research work.

Paper	Signal	Feature	Classification	Method	Accuracy	F1
[11]	WESAD dataset (Multi-signals)	GAF encoding	4-class	CNN	94.77%	95%
[13]	WESAD dataset (Multi-signals)	-	3-class	BNN + ANN	94%	96.9%
[27]	WESAD dataset (Multi-signals)	Time and Freq	2-class	ANN	95.21%	94.24%
[27]	WESAD dataset (Multi-signals)	Time and Freq	3-class	ANN	84.32%	78.71%
[37]	WESAD dataset (ECG only)	DCT Freq	3-class	X-GWO-SVM	95.93%	95.56%
[14]	RESP	Time and Freq, RQA, and approximate entropy	2-class	MLP (LOOCV)	94.4%	-
[16]	RESP	Time and Freq	3-class	SVM	93.41%	-
[17]	RESP	Time and Freq	3-class	KNN	92.06%	-
[18]	RESP	Time and Freq	4-class	SVM	92.5%	95.11%
Proposed work	WESAD dataset (RESP only)	Physiologically meaningful respiratory feature	3-class	Stacking	92.33%	92.32%

Unlike previous respiratory-based studies [14,16–18], which primarily rely on statistical features in time or frequency domains, our method emphasizes physiologically grounded indicators such as IT_ratio and RSBI. These physiologically meaningful features enhance interpretability, and when integrated within an interpretable modeling framework, our method achieves comparable performance to prior studies. This highlights the promise of physiologically interpretable approaches for effective stress recognition.

5.3. SHAP Value Analysis for Feature Mechanism

Figure 6 SHAP analysis reveals state-dependent importance of respiratory features, consistent with their physiological roles. Key stress-related metrics (e.g., BR, IT_ratio, D) exhibit significant contributions. Notably, ET_cv critically discriminates normal and stress states, showing opposite correlations, reflecting divergent respiratory dynamics. TE_cv strongly negatively correlates with the normal state but minimally affects meditation, while variability coefficients hold higher importance in stress versus meditation. These findings suggest distinct discriminative features across states: stress arousal detection in normal states, respiratory instability in stress, and prolonged and stable cycles in meditation.

Figure 8 shows RSBI_cv as the most influential predictor for Subject 16, while conventional metrics like BR underperform compared to BR_cv, IT_ratio, ET_cv, and RSBI_cv. This suggests that several physiologically relevant features, which have not been previously used in stress classification, can capture relevant information for stress state classification. SHAP value analysis explains the causal relationship between feature contribution and physiological states, confirming the feature physiology correlation via explainable machine learning. This gives a biological rationale to the classification model and shows respiratory parameters' potential as stress physiological indicators. Moreover, quantifying feature contributions may guide the optimization of respiratory metrics for personalized stress management.

5.4. Limitations and Future Directions

Though our model performed well in LOSO experiments, some individuals still had poor classification results after baseline removal. For example, Subject 13 had a classification accuracy of only 67.4%. This may be due to poor signal quality, making it hard to extract precise feature parameters and thus reducing classification accuracy. Moreover, the WESAD dataset employed in this research comprises data from only 15 participants. It is a relatively limited sample size. The pronounced gender imbalance (predominantly male subjects) in the dataset could introduce bias in certain respiratory features, potentially affecting the reliability of cross-individual testing. In addition, certain potential factors during data collection, such as time of day and environmental conditions, may also influence respiratory patterns [38]. Therefore, our conclusions are drawn within the specific context of the dataset and methodology employed in this study. Future work should optimize the feature set and expand the sample size to further explore the applicability of these features across broader populations.

Currently, the respiratory signals utilized are exclusively collected from the chest. Recent findings suggest that various breathing techniques, including thoracic and abdominal breathing, can serve as indicators of an individual's stress state [39,40]. In future work, we aspire to gather signals from both chest and abdominal breathing. Analyzing these two breathing types under different stress states can help us extract features for stress recognition and manage stress dynamically. This approach promises greater benefits for the evaluation of meditation intervention effects and the formulation of personalized strategies.

6. Conclusions

The study developed an interpretable stress classification framework using physiologically inspired respiratory features. To the best of our knowledge, this is the first work to apply such a framework to stress classification using respiratory signals. By integrating statistical analysis, a stacking ensemble model, and SHAP-based interpretability, the framework not only effectively classifies stress states but also identifies key respiratory features contributing to model decisions, thereby enhancing transparency. The stacking ensemble model achieved a LOSO cross-validation accuracy of 92.33% on the WESAD dataset, demonstrating comparable performance to existing models that rely on traditional statistical features or multimodal signals. Importantly, features such as BR, IT_ratio, and ET_cv were found to align with autonomic regulation mechanisms. RSBI also showed strong discriminative value. Notably, an increase in respiratory depth under stress was observed, suggesting a need for further investigation. These findings confirm that physiologically meaningful respiratory features alone can provide effective stress classification while enhancing model transparency.

Furthermore, the proposed framework quantitatively identifies key respiratory features relevant to stress detection, offering valuable insights for designing targeted, physiology-based interventions for stress management. And integrating these physiologically meaningful features and interpretable frameworks into multimodal approaches offers a promising direction for enhancing both the explainability and clinical applicability of stress monitoring systems. Future research will focus on personalized calibration algorithms and multimodal respiratory synergy analysis to enhance the clinical utility of stress management systems.

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